EC-390: Digital Signal Processing

LECTURE NO 2: *Discrete Time Signals and Systems*

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Organization of Lecture

1. Discrete and Digital Signals.
2. Representation of Discrete Time Signals.
3. Standard Discrete Time Signals.

1. Discrete and Digital Signals

- The ***discrete signal*** is a function of a discrete independent variable.
- The letter "n" is used to denote the independent variable. The discrete or digital signal is denoted by x(n). The discrete signal is defined for every integer value of the independent variable "n".
- When the independent variable is time t, the discrete signal is called ***discrete time signal***. In discrete time signal, the time is quantized uniformly using the relation t = nT, where T is the sampling time period. (The sampling time period is inverse of sampling frequency). The discrete time signal is denoted by x(n) or x(nT).
- The digital signal is same as discrete signal except that the magnitude of the signal is quantized. Quantization is necessary to represent the signal in binary codes.

2. Representation of Discrete Time Signals (1/4)

The discrete time signal can be represented by following methods:

1. Functional representation

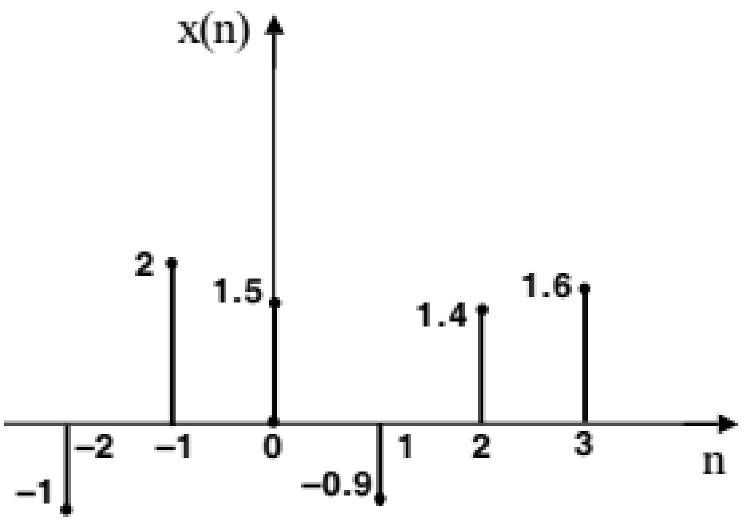
In functional representation the signal is represented as mathematical equation, as shown in the following example.

$$x(n) = \begin{cases} -1 & ; \quad n = -2 \\ 2 & ; \quad n = -1 \\ 1.5 & ; \quad n = 0 \\ -0.9 & ; \quad n = 1 \\ 1.4 & ; \quad n = 2 \\ 1.6 & ; \quad n = 3 \\ 0 & ; \quad \text{other } n \end{cases}$$

2. Representation of Discrete Time Signals (2/4)

2. Graphical representation

In graphical representation the signal is represented in a two dimensional plane. The independent variable is represented in the horizontal axis and the value of the signal is represented in the vertical axis as shown in fig.



2. Representation of Discrete Time Signals (3/4)

3. Tabular representation

In tabular representation, two rows of a table are used to represent a discrete time signal. In the first row the independent variable "n" is tabulated and in the second row the value of the signal for each value of "n" are tabulated as shown in the following example.

n	....	–2	–1	0	1	2	3	....
x(n)	....	–1	2	1.5	–0.9	1.4	1.6	....

2. Representation of Discrete Time Signals (4/4)

4. Sequence representation

In sequence representation, the discrete time signal is represented as one dimensional array as shown in the following examples.

An infinite duration discrete time signal with the time origin, $n = 0$, indicated by the symbol $\uparrow$ is represented as,

$$x(n) = \{\ \ldots\ldots\ -1,\ 2,\ 1.5,\ -0.9,\ \underset{\uparrow}{1.4},\ 1.6,\ \ldots\ldots\ \}$$

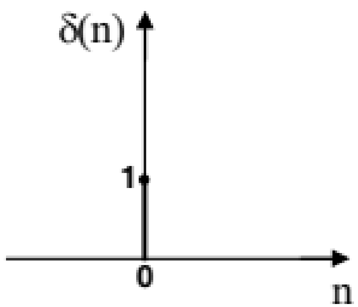
A finite duration discrete time signal with the time origin, $n = 0$, indicated by the symbol $\uparrow$ is represented as,

$$x(n) = \{\ -1,\ 2,\ \underset{\uparrow}{1.5},\ -0.9,\ 1.4,\ 1.6\ \}$$

3. Standard Discrete Time Signals (1/6)

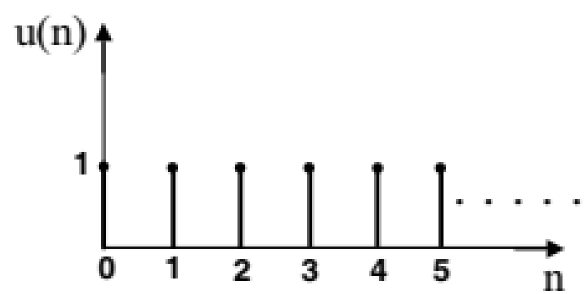
1. Digital impulse signal or Unit sample sequence

$$\text{Impulse signal, } \delta(n) = 1 \quad ; \quad n = 0$$
$$= 0 \quad ; \quad n \neq 0$$



2. Unit step signal

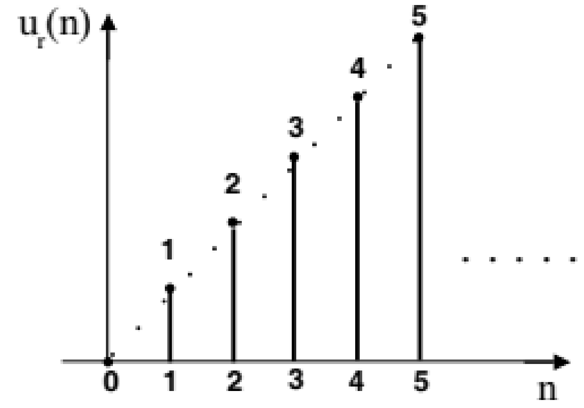
$$\text{Unit step signal, } u(n) = 1 \; ; \; n \geq 0$$
$$= 0 \; ; \; n < 0$$



3. Standard Discrete Time Signals (2/6)

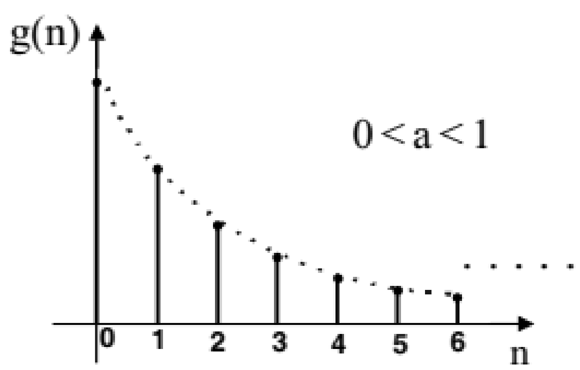
3. Ramp signal

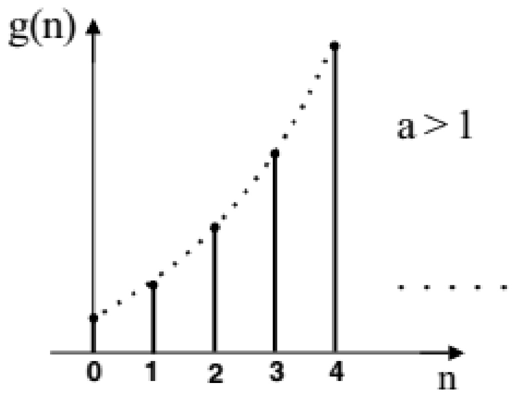
$$\text{Ramp signal, } u_r(n) = n \;;\; n \geq 0$$
$$= 0 \;;\; n < 0$$



4. Exponential signal

$$\text{Exponential signal, } g(n) = a^n \;;\; n \geq 0$$
$$= 0 \;;\; n < 0$$





3. Standard Discrete Time Signals (3/6)

5. Discrete time sinusoidal signal

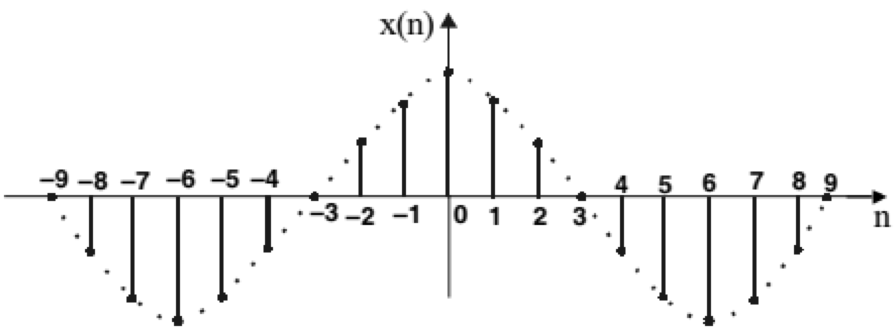
The discrete time sinusoidal signal may be expressed as,

$$x(n) = A\cos(\omega_0 n + \theta) \; ; \text{ for n in the range } -\infty < n < +\infty$$

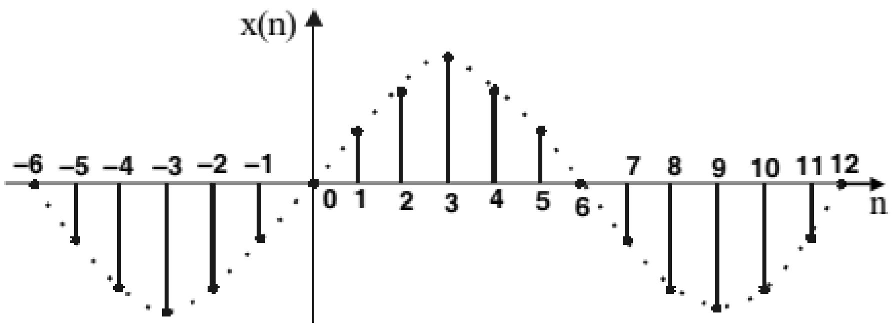
$$x(n) = A\sin(\omega_0 n + \theta) \; ; \text{ for n in the range } -\infty < n < +\infty$$

where, ω_0 = Frequency in radians/sample ; θ = Phase in radians

$$f_0 = \frac{\omega_0}{2\pi} = \text{Frequency in cycles/sample}$$



Discrete time sinusoidal signal represented by equation $x(n) = A\cos(\omega_0 n)$.



Discrete time sinusoidal signal represented by equation $x(n) = A\sin(\omega_0 n)$.

3. Standard Discrete Time Signals (4/6)

Properties of Discrete Time Sinusoid

1. A discrete time sinusoid is periodic only if its frequency f is a rational number, (i.e., ratio of two integers).
2. Discrete time sinusoids whose frequencies are separated by integer multiples of 2π are identical.

$$\therefore x(n) = A\cos[(\omega_0 + 2\pi k)n + \theta], \qquad \text{for } k = 0,1,2...$$

Proof :

$$\cos[(\omega_0 + 2\pi k)n + \theta] = \cos(\omega_0 n + 2\pi nk + \theta) = \cos[(\omega_0 n + \theta) + 2\pi nk]$$

$$= \cos(\omega_0 n + \theta)\cos 2\pi nk - \sin(\omega_0 n + \theta)\sin 2\pi nk$$

Since n and k are integers, $\cos 2\pi nk = 1$ & $\sin 2\pi nk = 0$

$$\therefore \cos[(\omega_0 + 2\pi k)n + \theta] = \cos(\omega_0 n + \theta), \qquad \text{for } k = 1, 2, 3, ...$$

3. Standard Discrete Time Signals (5/6)

6. Discrete time complex exponential signal

The discrete time complex exponential signal is defined as,

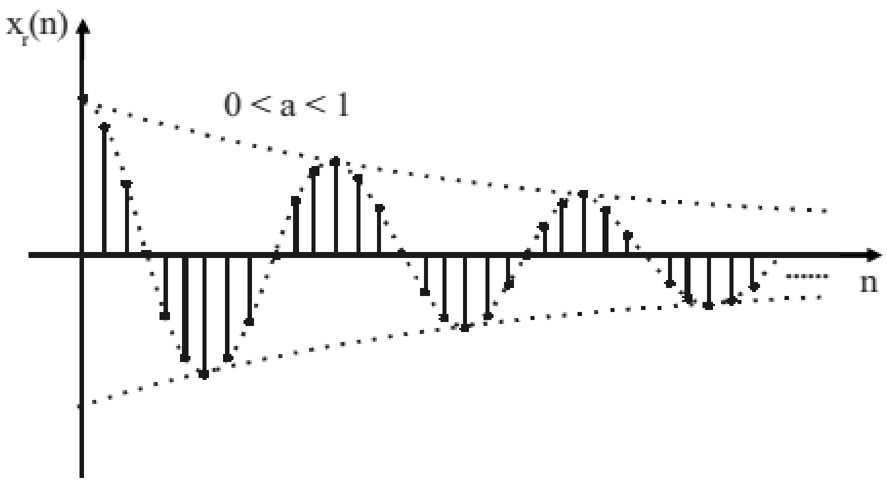
$$x(n) = a^n e^{j(\omega_0 n + \theta)} = a^n [\cos(\omega_0 n + \theta) + j \sin(\omega_0 n + \theta)]$$

$$= a^n \cos(\omega_0 n + \theta) + j\, a^n \sin(\omega_0 n + \theta) = x_r(n) + j\, x_i(n)$$

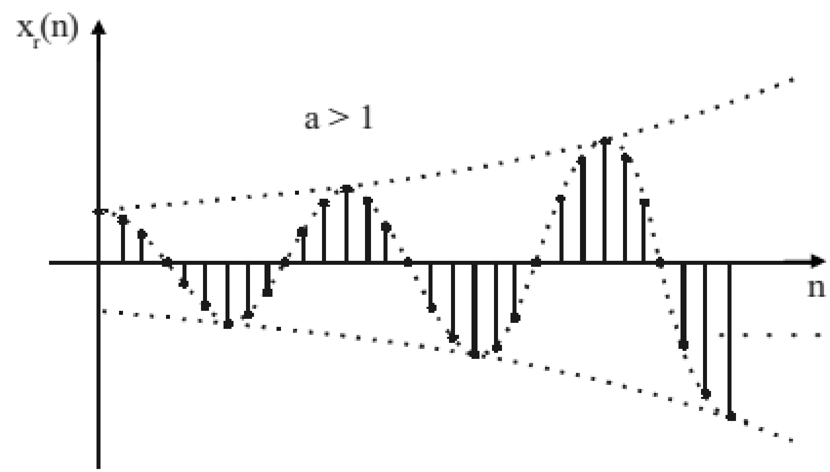
where, $x_r(n) = \text{Real part of } x(n) = a^n \cos(\omega_0 n + \theta)$

$x_i(n) = \text{Imaginary part of } x(n) = a^n \sin(\omega_0 n + \theta)$

The real part of x(n) will give an exponentially increasing cosinusoid sequence for $a > 1$ and exponentially decreasing cosinusoid sequence for $0 < a < 1$.



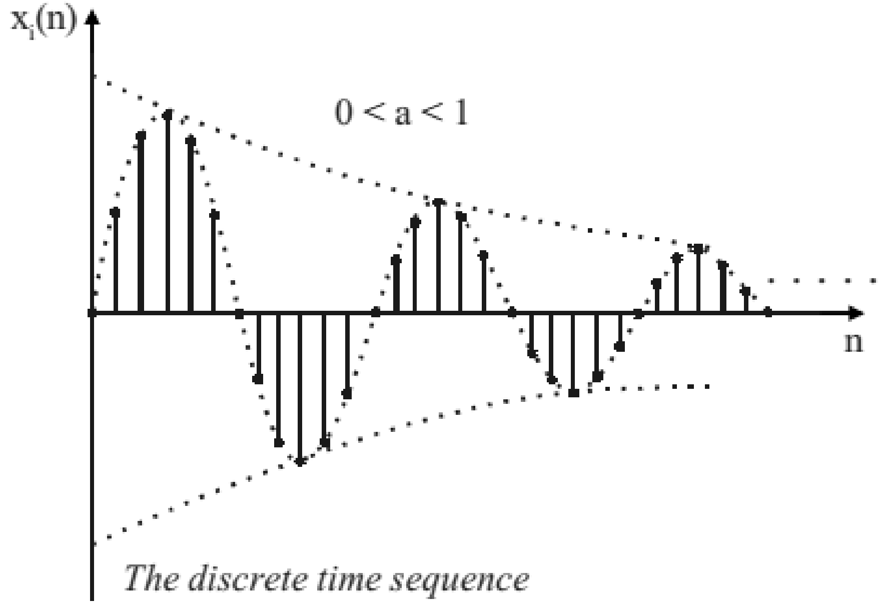
The discrete time sequence represented by the equation, $x_r(n) = a_n \cos \omega_0 n$ for $0 < a < 1$.



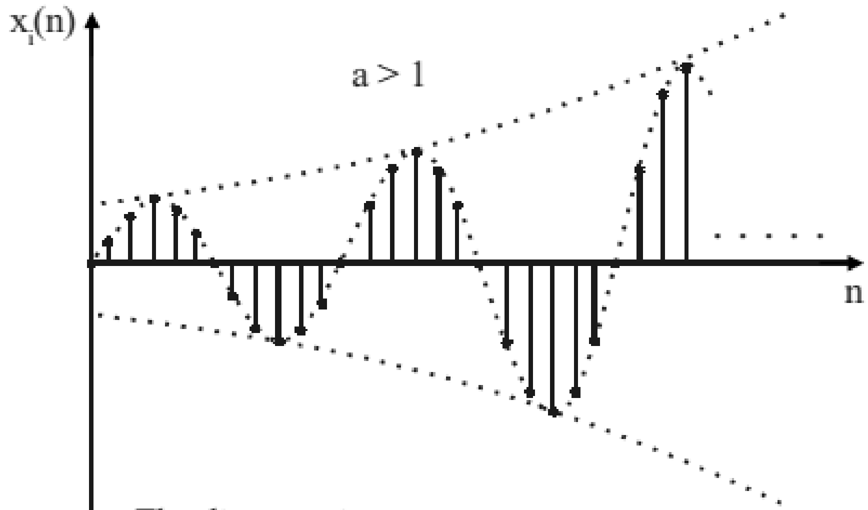
The discrete time sequence represented by the equation $x_r(n) = a_n \cos \omega_0 n$ for $a > 1$.

3. Standard Discrete Time Signals (6/6)

The imaginary part of x(n) will give rise to an exponentially increasing sinusoid sequence for a > 1 and exponentially decreasing sinusoid sequence for 0 < a < 1.



The discrete time sequence represented by the equation, $x_i(n) = a^n \sin \omega_0 n$ for $0 < a < 1$.



The discrete time sequence represented by the equation $x_i(n) = a^n \sin \omega_0 n$ for $a > 1$.

References: Textbooks

1. *“Discrete-Time Signal Processing”, International Edition, by Oppenheim, Alan V; Schafer, Ronald W, (2013), Pearson.*
2. *“Digital Signal Processing Using Matlab”, 4th Edition by Vinay K. Ingle, John G. Proakis, (2016), Cengage Learning.*
3. *Related material from open source, mainly internet.*